

Project Specification: Industrial Monitoring System

1. Project Theme

Concept

An industrial monitoring system that analyzes vibration, pressure, and temperature sensor data to enable predictive maintenance of machinery.

Data Acquisition Layer

Virtual or physical sensors that continuously collect:

- 1. Vibration (Hz)*
- 2. Pressure (bar)*
- 3. Temperature (°C)*

Data Publishing

- 1. Each sensor has a site gateway acting as a local aggregator.*
- 2. The telemetry data is sent to the cloud using a bandwidth-savvy protocol.*

Processing & Business Logic

- 1. Hazard conditions are detected when telemetry reveals operating thresholds are compromised.*
- 2. People in charge get that information on their UI.*

Integration API

Exposes endpoints to programmatically list industrial assets:

- 1. Location*
- 2. Model*
- 3. Specifications*

Provides access to telemetry readings:

- 4. *Current status*
 - 5. *Historical data intervals*
- Enables external integration with other systems.*

Web Portal

Web-based dashboard for real-time tracking of:

- 1. *Machine health*
 - 2. *Operational conditions*
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2. Personas

João Neves — Maintenance Technician



Age: 26

Occupation: Maintenance Technician

Location: Lisbon, Portugal

Background

João is a Maintenance Technician at SmartSense and is relatively new at the company. He is responsible for checking systems and equipment, making sure everything runs smoothly. João enjoys understanding how each machine works and values an organized and safe work environment.

Daily Life

In his daily life, João performs regular inspections and tests on equipment, logs any problems, and tries to solve them independently before involving the Maintenance Director. He uses manuals, digital tools, and monitoring software to diagnose failures. Even when faced with unexpected problems, João always tries to stay calm and ensure the company's systems run smoothly.

Goals & Needs

João wants to get promoted to Maintenance Director and increase his salary to support his family. He also wants to improve his technical skills and learn new maintenance technologies to stand out in the company. For him, career growth means recognition and financial security.

Motivation

What drives João is the desire to be recognized for his effort and work, as well as to have professional stability. He wants to feel that his contributions truly matter and that his growth is valued by the company.

Manuel Gomes — Maintenance Director



Age: 34

Occupation: Maintenance Director

Location: Lisbon, Portugal

Background

Manuel has been working at SmartSense for many years and knows all the company's systems and equipment inside out. He leads the maintenance team and is responsible for ensuring that all sensors, machines, and systems run smoothly. Manuel values organized processes, operational safety, and team efficiency.

Daily Life

In his daily life, Manuel reviews the full history of failures, identifies failure patterns, and assesses the need for more critical actions to prevent bigger issues. He guides technicians like João and provides constant feedback. His work requires attention to detail and the ability to anticipate problems before they affect company operations.

Goals & Needs

Manuel wants to keep the company's operations efficient and safe, ensuring the team works in a coordinated way. He seeks ways to reduce failures and maintenance costs, while also investing in the technical development of his team.

Motivation

What motivates Manuel is the trust the company places in him and try turn the company better. He values professional recognition, a good team and efficeience.

Sara Lopes — Administrator



Age: 32

Occupation: Administrator

Location: Lisbon, Portugal

Background

Sara works at SmartSense as an Administrator responsible for overseeing operational efficiency, team organization, and equipment management. She has extensive experience in operations and team coordination, ensuring that both technical teams and company resources are used effectively.

Daily Life

In her daily work, Sara reviews machine performance reports and operational data. She monitors indicators such as operating time, failures and maintenance interventions. She also checks whether technicians are completing their assigned tasks on time and ensures that all machines and equipment are properly registered in the system. Sara frequently communicates with the Maintenance Director and technical teams to make sure operations run smoothly and everyone is aligned.

Goals & Needs

Sara wants to maintain an efficient and well-organized operation. She needs clear information about machine performance, maintenance activities, and team productivity in order to decide whether equipment should continue operating or be replaced.

Motivation

Sara is motivated by improving operational efficiency and helping the company grow while maintaining high quality standards. She enjoys seeing teams working productively, processes running smoothly, and the organization operating in a clear and structured way.

3. User Stories

User Story 1:

As a Maintenance Technician,
I want to know if there is a machine with a breakdown,
so that I can fix it.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the user is logged in the system
And the user has the role of Maintenance Technician
When they navigate to the 'Machines' tab
Then a tab labeled 'Machines' should be visible.

User Story 2:

As a Maintenance Technician,
I want to monitor early warning signs from the machines,
so that I can perform preventive repairs before a total breakdown occurs.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the user is logged into the system
And the user has the role of "Maintenance Technician"
When the user is viewing a machine's Individual Dashboard
Then a section labeled "Health Status" should be visible
And the section should display:

1. A vibration trend graph (time vs value)
 2. A pressure trend graph (time vs value)
 3. A temperature trend graph (time vs value)
- And each graph should clearly label the time axis and the value axis
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User Story 3:

As a Maintenance Technician,
I want to view each machine's priority level based on its importance, downtime, and fault severity,
so that I can optimize my workflow and fix the most critical issues first.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the user is logged into the system

When the user navigates to the machine overview or ranking section

Then the system should display a ranking of machines

And the ranking should determine the priority order of the machines

User Story 4:

As a Maintenance Technician,

I want to request assistance for a machine in the app when a repair requires additional help,

so that my colleagues are notified and can assist me.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the Maintenance Technician has completed the assistance request form

When the Technician submits the request

Then the system should notify the "Maintenance Director".

User Story 5:

As a Maintenance Director,

I want to know if there is a machine that has been having several breakdowns,

so that I can investigate a more serious fault.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the user is logged in the system

And the user has the role of Maintenance Director

When they navigate to the 'Machines' tab

Then a section labeled 'History' should be visible, where information regarding the machine should be visible, such as title, description and previous breakdowns ordered by their date.

User Story 6:

As a Maintenance Director,

I want to choose the technicians that can help when there is a request sent by another technician,

so that the machine can be fixed quicker.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the user is logged in the system

And the user has the role of Maintenance Director

When they navigate to the 'Requests' tab

Then a section labeled 'Requests' should be visible, where information regarding the request, such as

1. description
2. name of the technician who asked for help
3. location
4. machine's name

And a button/dropdown for the director assign people, for each request

User Story 7:

As an Administrator,

I want to monitor the technicians' performance,

so that I can ensure the maintenance team is meeting productivity expectations.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the user is logged into the system

And the user has the role of "Administrator"

When the user navigates to the main interface

Then a tab labeled "Team" should be visible

User Story 8:

As a Administrator,
I want to register new equipment/machines in the app,
so that the maintenance team can start tracking its performance and history.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the user is logged into the system
And the user has the role of "Administrator"
When the user navigates to the main interface
Then a tab labeled "Managing" should be visible

User Story 9:

As a Administrator,
I want to remove equipment/machines from the app,
so that the views of every user remain updated.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the user is logged in as a Administrator
And the equipment exists in the system
When the user clicks the "Delete" button for that equipment
And confirms the deletion
Then the equipment is removed from all user views immediately
And a success message "Equipment deleted successfully" is displayed
And the deletion is logged in the system audit trail

User Story 10:

As a Administrator,
I want to access the history of removed machines,
so that I can review past performance, costs, and maintenance records for several purposes.

Description: No details added (to be added as the project develops).

Acceptance Criteria:

Given the user is logged into the system
And the user has the role of “Administrator”
When the user navigates to the main interface
Then a section labeled “Archived” should be visible
And this section should provide access to removed or deactivated machines

Scenario 1 — User Authentication

Actors:

User (Administrator, Maintenance Director, Maintenance Technician)

Description:

A user logs into the system to access functionalities based on their role.

Steps:

1. The user enters their username and password.
2. The system verifies the credentials.
3. The system identifies the user role.
4. The system gives access to functionalities based on the role.

Relationships:

Inheritance:

Administrator, Maintenance Director, and Maintenance Technician inherit from User

Dependency:

Authentication system depends on the User credentials and rules

Scenario 2 — Registering a Machine

Actors:

Administrator

Description:

The administrator registers a new machine in the system.

Steps:

1. The administrator accesses the machine management module.
2. The administrator inputs machine details (ID, name, location, sensors).
3. The system validates the information.
4. The system stores the machine in the database.

Relationships:

Association:

A Machine is associated with multiple Sensors.

Dependency:

The system depends on the Machine entity to store information.

Scenario 3 — Sensor Information Collection

Actors:

System (Sensors)

Description:

The sensors collect information from the machines.

Steps:

1. The sensors measure vibration, pressure, and temperature.
2. The information is sent to the system.
3. The system stores the information in the database.
4. The information becomes available for monitoring and analysis.

Relationships:

Association:

A Machine is associated with multiple Sensors.

Dependency:

The Failure Manager depends on sensor data.

Scenario 4 — Error Detection

Actors:

Error Manager (System)

Description:

The system analyzes sensor information to detect errors.

Steps:

1. The system analyzes the latest sensor information.
2. The system detects strange patterns.
3. The system gives an error.
4. The error is stored in the system logs.
5. The error is made visible to the Maintenance Director.

Relationships:

Dependency:

Error Manager depends on Sensor data.

Association:

An error is associated with a Machine.

Scenario 5 — Assigning Maintenance

Actors:

Maintenance Director

Description:

The Maintenance Director gives a maintenance task to a maintenance technician.

Steps:

1. The Maintenance Director reviews the error notification.
2. The director selects a maintenance technician.
3. The system creates a maintenance notification.
4. The technician is notified.

Relationships:

Association:

Maintenance is associated with a Machine and a Maintenance Technician.

Inheritance:

Normal Maintenance and Special Maintenance inherit from Maintenance.

Scenario 6 — Maintenance Technician Sends a Request

Actors:

Maintenance Technician

Description:

A maintenance technician sends a request for additional help during the maintenance.

Steps:

1. The maintenance technician accesses the task.
2. The maintenance technician creates a request asking for help and report additional i
3. The system stores the request.
4. The maintenance director receives the request.

Relationships:

Association:

The Request is associated with a Maintenance Technician.

Dependency:

The Maintenance Director depends on the request information.

Scenario 7 — Monitoring Machines

Actors:

Maintenance Technician, Maintenance Director, Administrator

Description:

Users monitor the machines and their information.

Steps:

1. The user accesses the system interface.
2. The system displays their health status.
3. The system displays the sensors information:
 - Vibration
 - Pressure
 - Temperature
4. The user analyzes the machine data.

Relationships:**Association:**

Machine is associated with Sensors.

Dependency:

Monitoring depends on sensor data.

Scenario 8 — Removing a Machine

Actors:

Administrator

Description:

The administrator removes a machine from the system.

Steps:

1. The administrator selects a machine.
2. The administrator clicks in the delete option.
3. The system asks for confirmation.
4. The administrator confirms the deletion.
5. The system removes the machine.
6. The system displays a success message.
7. The action is logged in the system.

Relationships:

Dependency:

The system depends on the Machine entity.

Dependency:

The system logs user actions.

Scenario 9 — Viewing Archived Machines

Actors:

Administrator

Description:

The administrator accesses removed machines.

Steps:

1. The administrator navigates to the "Archived" section
2. The system displays all the removed machines
3. The administrator reviews the historical data and the reasons.

Relationships:

Association:

Machine is associated with a historical data

Dependency:

The system depends on stored information

4. Architecture

 UML Class Diagram

5. System Requirements

Functional Requirements

- **FR1:** The system shall support different types of users.

- **FR2:** The system shall authenticate users before allowing access to system functionalities.
- **FR3:** The system shall restrict and allow access to certain functionalities based on the user type.
- **FR4:** The system shall store machine information (e.g., identification, status, and operational data).
- **FR5:** The system shall allow machine information to be updated.
- **FR6:** The system shall collect data from vibration, pressure, and temperature sensors.
- **FR7:** The system shall store historical sensor data for future analysis.
- **FR8:** The system shall automatically generate an issue record when a failure is detected.
- **FR9:** The system shall record maintenance activities performed and those that are still pending.
- **FR9:** The system shall allow maintenance technicians to send requests.

Non-Functional Requirements

- **NFR1:** The system shall process sensor data in near real time.
 - **NFR2:** The system shall support the monitoring of multiple machines simultaneously.
 - **NFR3:** The system shall ensure the integrity of data collected from sensors.
 - **NFR4:** The system shall ensure secure user authentication.
 - **NFR5:** The system shall log user activities.
 - **NFR6:** The system interface shall be simple and intuitive for technicians and managers.
 - **NFR7:** The system shall allow new machines and sensors to be added without significant performance degradation.
 - **NFR8:** The system shall allow future integration with new monitoring technologies.
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6. System Entities and Domain Structure

6.1. User (Base Class)

All roles inherit from this base class.

Field	Type	Description
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id	UUID	Unique identifier
name	String	Full name
email	String	Login and notifications
passwordHash	String	Hashed password
gender	enum	MALE/FEMALE/OTHER/PREFERNOTTOSAY
phoneNumber	String	Optional
createdAt	datetime	Account creation timestamp
isPrivileged	boolean	Admin-like privileges

Subclasses:

- Technician
- MaintenanceTechnician
- MaintenanceDirector
- Admin

6.2. Technician

Represents a maintenance technician with performance metrics.

Field	Type	Description
numberOfFaultsFixed	int	Number of faults the technician has fixed
assistedCounter	int	Number of times the technician assisted another
wasAssistedCounter	int	Number of times the technician required help
averageRepairTime	double	Average repair time for tasks
tasksCompleted	int	Total completed tasks
tasksPending	int	Tasks still pending
isAvailable	boolean	Whether the technician is available
currentAssignment	Machine	Machine currently assigned

skillSet	List	Skills and specializations
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6.3. Maintenance Technician

A specialized technician for maintenance tasks.

Inherits from `User` and is used specifically for maintenance operations.

Field	Type	Description
(inherits all fields from User)	—	—
—	—	No additional fields beyond User

6.4. Maintenance Director

Field	Type	Description
technicianIds	List	IDs of technicians managed
machineIds	List	IDs of machines managed

Inherits from `User`.

6.5. Machine

Field	Type	Description
machineId	UUID	Unique identifier
name	String	Machine name
location	String	Physical location
importanceLevel	int	Importance for prioritization
status	enum	ACTIVE / INACTIVE / MAINTENANCE / BROKEN
createdAt	datetime	Registration timestamp
suspicionFlag	boolean	Indicates suspected malfunction

sensors	List	Sensors associated with the machine
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6.6. Maintenance

Field	Type	Description
maintenanceId	UUID	Unique identifier
machine	Machine	Machine under maintenance
technician	MaintenanceTechnician	Technician assigned
type	enum	NORMAL / URGENT
status	enum	PENDING / IN_PROGRESS / COMPLETED
notes	String	Optional notes
startTime	datetime	When maintenance started
endTime	datetime	When maintenance ended

6.7. Problem

Field	Type	Description
problemID	UUID	Unique identifier
machine	Machine	Machine where the problem occurred
description	String	Description of the issue
importanceLevel	int	Importance of the problem
priority	float	Calculated priority
status	enum	PENDING / IN_PROGRESS / RESOLVED
detectedAt	datetime	When the problem was detected
startProblemDate	datetime	When work on the problem started
solvedProblemDate	datetime	When the problem was resolved

resolved	boolean	Whether the problem is resolved
assignedTechnician	MaintenanceTechnician	Technician assigned
faultSeverity	String	Severity of the fault

6.8. Assistance Request

Used for **technician-to-technician assistance**.

Field	Type	Description
id	Long	Unique identifier
problem	Problem	Problem requiring assistance
requestedBy	Technician	Technician who requested help
reason	String	Reason for the request
status	enum	PENDING / ACCEPTED / COMPLETED
assignedTechnician	Technician	Technician assigned to assist
createdAt	datetime	Timestamp of request creation

6.9. Request (Problem Assistance)

Used for **requests associated with problems**, not technician-to-technician.

Field	Type	Description
requestID	UUID	Unique identifier
problem	Problem	Problem associated
requestedBy	User	User who created the request
reason	String	Reason for the request
assignedTechnician	MaintenanceTechnician	Technician assigned
status	enum	PENDING / ACCEPTED / COMPLETED

createdAt	datetime	Timestamp of creation
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6.10. Sensor & Readings

Sensor (abstract)

Field	Type	Description
id	UUID	Unique identifier
machine	Machine	Machine where the sensor is installed
readings	List	Historical readings

Reading (abstract)

Field	Type	Description
id	UUID	Unique identifier
timestamp	datetime	When the reading was taken
sensor	Sensor	Sensor that produced the reading

Specialized Readings

Type	Extra Field	Description
TemperatureReading	temperature	Temperature in °C
PressureReading	pressure	Pressure in bar

SensorReading (simplified model used by API)

Field	Type	Description
id	Long	Unique identifier
machine	Machine	Machine associated
sensorType	enum	TEMPERATURE / PRESSURE / VIBRATION

value	Double	Reading value
recordedAt	datetime	Timestamp

6.11. Relationships

User -> Technician / MaintenanceTechnician / MaintenanceDirector / Admin (inheritance)

Machine -> Problem (1 to many)

Machine -> Maintenance (1 to many)

Problem -> Request (1 to many)

Problem -> AssistanceRequest (1 to many)

MaintenanceTechnician -> Maintenance (1 to many)

Technician -> AssistanceRequest (1 to many, as requester or assigned helper)

7. Controllers & API Endpoints

AuthController

- POST /auth/login
- POST /auth/register

UserController

- GET /users
- GET /users/performance

MachineController

- GET /machines
- GET /machines/{id}

- POST /machines

ProblemController

- GET /problems
- POST /problems
- POST /problems/{id}/start
- POST /problems/{id}/complete

MaintenanceController

- POST /maintenance/assign
- POST /maintenance/start
- POST /maintenance/complete

AssistanceRequestController

- GET /api/v1/requests
- GET /api/v1/requests/{id}
- GET /api/v1/requests/technician/{technicianId}
- POST /api/v1/requests
- PUT /api/v1/requests/{id}
- PATCH /api/v1/requests/{id}/accept
- DELETE /api/v1/requests/{id}

RequestController

- POST /requests
- POST /requests/{id}/assign
- POST /requests/{id}/complete
- GET /requests
- PUT /requests/{id}/status

SensorController

- GET /api/v1/sensors/{machineId}
- GET /api/v1/sensors/{machineId}/{type}
- GET /api/v1/sensors/{machineId}/{type}/latest

- POST /api/v1/sensors
 - DELETE /api/v1/sensors/{id}
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8. Component Layer

The backend is structured into:

- **Domain Layer**
- **Repository Layer**
- **Service Layer**
- **Controller Layer**

Controllers include:

- AuthController
 - UserController
 - MachineController
 - ProblemController
 - MaintenanceController
 - AssistanceRequestController
 - RequestController
 - SensorController
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9. UML Class Diagram



UML Class Diagram

10. Conclusion

This report reflects the **current backend implementation**, aligned with:

- Entities
- Controllers
- Services
- API endpoints
- Architecture

- Requirements